

1.Task-1:

```
vennela_07@VennelaPC:~$ mkdir week3
vennela_07@VennelaPC:~$ cd week3
vennela_07@VennelaPC:~/week3$ nano task1.c
vennela_07@VennelaPC:~/week3$ gcc task.c -o task1
cc1: fatal error: task.c: No such file or directory
compilation terminated.
vennela_07@VennelaPC:~/week3$ gcc task1.c -o task1
vennela_07@VennelaPC:~/week3$ ./task1
myshell$ hello
You entered: hello
myshell$ abc
You entered: abc
myshell$ exit
Shell exited.
```

1.Task-2:

```
vennela_07@VennelaPC:~$ cd week3
vennela_07@VennelaPC:~/week3$ nano File1.txt
vennela_07@VennelaPC:~/week3$ cat File1.txt
This is File1.
This content will be copied to File2.
vennela_07@VennelaPC:~/week3$ nano task2.c
vennela_07@VennelaPC:~/week3$ gcc task2.c -o task2
vennela_07@VennelaPC:~/week3$ ./task2
File copied successfully.
vennela_07@VennelaPC:~/week3$ cat File2.txt
This is File1.
This content will be copied to File2.
```

```
vennela_07@VennelaPC:~/week3$ cd ~/week3
vennela_07@VennelaPC:~/week3$ cat File1.txt
This is File1.
This content will be copied to File2.
vennela_07@VennelaPC:~/week3$ nano task2.c
vennela_07@VennelaPC:~/week3$ cat task2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd1, fd2;
    char buffer[100];
    int n;

    fd1 = open("File1.txt", O_RDONLY);

    fd2 = open("File2.txt", O_CREAT | O_WRONLY, 0644)

    n = read(fd1, buffer, 100);

    write(fd2, buffer, n);

    close(fd1);
    close(fd2);

    printf("File copied successfully.\n");

    return 0;
}
vennela_07@VennelaPC:~/week3$ gcc task2.c -o task2
vennela_07@VennelaPC:~/week3$ ./task2
File copied successfully.
vennela_07@VennelaPC:~/week3$ cat File2.txt
This is File1.
This content will be copied to File2.
vennela_07@VennelaPC:~/week3$ ./task2
File copied successfully.
vennela_07@VennelaPC:~/week3$ cat File2.txt
This is File1.
This content will be copied to File2.
vennela_07@VennelaPC:~/week3$ |
```

```
GNU nano 7.2                                File1.txt
This is File1.
This content will be copied to File2.

GNU nano 7.2                                task2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd1, fd2;
    char buffer[100];
    int n;

    fd1 = open("File1.txt", O_RDONLY);

    fd2 = open("File2.txt", O_CREAT | O_WRONLY, 0644);

    n = read(fd1, buffer, 100);

    write(fd2, buffer, n);

    close(fd1);
    close(fd2);

    printf("File copied successfully.\n");

    return 0;
}
```

TASK3:

```
vennela_07@VennelaPC:~/week3$ gcc task2.c -o task2
vennela_07@VennelaPC:~/week3$ ./task2
File copied successfully.
vennela_07@VennelaPC:~/week3$ cat File2.txt
This is File1.
This content will be copied to File2.
vennela_07@VennelaPC:~/week3$ ./task2
File copied successfully.
vennela_07@VennelaPC:~/week3$ cat File2.txt
This is File1.
This content will be copied to File2.
vennela_07@VennelaPC:~/week3$ nano task3.c
vennela_07@VennelaPC:~/week3$ gcc task3.c -o task3
vennela_07@VennelaPC:~/week3$ ./task3
Enter command: hello
You entered: hello
```

GNU nano 7.2

task3.c

```
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
|
int main()
{
    struct termios oldt, newt;
    char buffer[100];
    int position = 0;
    char ch;

    tcgetattr(STDIN_FILENO, &oldt);

    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    printf("Enter command: ");
    fflush(stdout);
```

```

while (1)
{
    read(STDIN_FILENO, &ch, 1);

    /* Enter key */
    if (ch == '\n' || ch == '\r')
    {
        buffer[position] = '\0';

        printf("\nYou entered: %s\n", buffer);

        if (position == 4 &&
            buffer[0] == 'e' &&
            buffer[1] == 'x' &&
            buffer[2] == 'i' &&
            buffer[3] == 't')
        {
            break;
        }
    }
}

```

TASK 4:

```

vennela_07@VennelaPC:~/week3$ cd ~/week3
vennela_07@VennelaPC:~/week3$ nano task4.c
vennela_07@VennelaPC:~/week3$ gcc task4.c -o task4
vennela_07@VennelaPC:~/week3$ ./task4
myshell> hello
myshell> linux
myshell> gcc
myshell> |

```

```

#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define MAX 100
#define HISTORY_SIZE 10

int main()
{
    char buffer[MAX];
    char history[HISTORY_SIZE][MAX];
    char ch;
    int index = 0;
    int history_count = 0;
    int history_pos = 0;

    struct termios oldt, newt;

    tcgetattr(STDIN_FILENO, &oldt);

    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while(1)
    {
        write(STDOUT_FILENO, "mysh$ ", 6);

```

```

        index = 0;
        history_pos = history_count;

        while(1)
        {
            read(STDIN_FILENO, &ch, 1);

            if(ch == '\n' || ch == '\r')
            {
                buffer[index] = '\0';
                write(STDOUT_FILENO, "\n", 1);

                if(strcmp(buffer, "exit") == 0)
                {
                    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
                    return 0;
                }

                if(index > 0)
                {
                    strcpy(history[history_count % HISTORY_SIZE], buffer);

                    if(history_count < HISTORY_SIZE)
                        history_count++;

                    history_pos = history_count;

```

```

        printf("Command: %s\n", buffer);
        break;
    }

    else if(ch == 127 || ch == 8)
    {
        if(index > 0)
        {
            index--;
            write(STDOUT_FILENO, "\b \b", 3);
        }
    }

    else if(ch == 27)
    {
        char seq[2];

        read(STDIN_FILENO, &seq[0], 1);
        read(STDIN_FILENO, &seq[1], 1);

        if(seq[0] == '[' && seq[1] == 'A')
        {
            if(history_count > 0 && history_pos > 0)
            {
                history_pos--;

                strcpy(buffer, history[history_pos]);
            }
        }
    }

```

```

        index = strlen(buffer);

        write(STDOUT_FILENO, "\r\033[K", 4);
        write(STDOUT_FILENO, "mysh$ ", 6);
        write(STDOUT_FILENO, buffer, index);
    }
}

else if(seq[0] == '[' && seq[1] == 'B')
{
    if(history_pos < history_count - 1)
    {
        history_pos++;

        strcpy(buffer, history[history_pos]);

        index = strlen(buffer);

        write(STDOUT_FILENO, "\r\033[K", 4);
        write(STDOUT_FILENO, "mysh$ ", 6);
        write(STDOUT_FILENO, buffer, index);
    }
    else
    {
        history_pos = history_count;
        index = 0;
        buffer[0] = '\0';
    }
}

```

```

        history_pos = history_count;
        index = 0;
        buffer[0] = '\0';

        write(STDOUT_FILENO, "\r\033[K", 4);
        write(STDOUT_FILENO, "mysh$ ", 6);
    }
}

else
{
    if(index < MAX - 1)
    {
        buffer[index] = ch;
        index++;

        write(STDOUT_FILENO, &ch, 1);
    }
}

}

tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

return 0;
}

```

TASK 5:

```
vennela_07@VennelaPC:~/week3$ nano task5.c
vennela_07@VennelaPC:~/week3$ gcc -Wall -Wextra task5.c -o task5
vennela_07@VennelaPC:~/week3$ ./task5
Enter command: hello
Stored: hello
Enter command: linux
Stored: linux
Enter command: gcc
Buffer resized to 4.
Stored: gcc
Enter command: ls
Stored: ls
Enter command: pwd
Buffer resized to 8.
Stored: pwd
Enter command: exit

Commands stored:
1. hello
2. linux
3. gcc
4. ls
5. pwd

Memory released successfully.
vennela_07@VennelaPC:~/week3$ valgrind --version
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.22.0-0ubuntu2
See 'snap info valgrind' for additional versions.
vennela_07@VennelaPC:~/week3$ sudo apt update
[sudo] password for vennela_07:
Get:1 https://deb.nodesource.com/node_22.x nodistro InRelease [12.1 kB]
Get:2 https://deb.nodesource.com/node_22.x nodistro/main amd64 Packages [12.0 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [934 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1191 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [202 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [282 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.3 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1201 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1683 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [335 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [240 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.2 kB]
```

```

Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1424 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1330 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [323 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [304 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [10.9 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.6 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5744 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [31.0 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 10.6 MB in 7s (1531 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
209 packages can be upgraded. Run 'apt list --upgradable' to see them.
vennela_07@VennelaPC:~/week3$ sudo apt install valgrind
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libdw1t64
  libelf1t64 libipt2 libsource-highlight-common libsource-highlight4t64
Suggested packages:
  gdb-doc gdbserver valgrind-dbg valgrind-mpi kcachegrind alleyoop valkyrie
The following NEW packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2
  libsource-highlight-common libsource-highlight4t64 valgrind
The following packages will be upgraded:
  libdw1t64 libelf1t64
2 upgraded, 9 newly installed, 0 to remove and 207 not upgraded.
Need to get 27.0 MB/27.3 MB of archives.
After this operation, 103 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-common all 0.190-1.1ubuntu0.1 [14.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libbabeltrace1 amd64 1.5.11-3build3 [164 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod1t64 amd64 0.190-1.1ubuntu0.1 [17.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libipt2 amd64 2.0.6-1build1 [45.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight-common all 3.1.9-4.3build1 [64.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build1 [258 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gdb amd64 15.1-1ubuntu1~24.04.1 [4083 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libc6-dbg amd64 2.39-0ubuntu8.8 [7461 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 valgrind amd64 1:3.22.0-0ubuntu3 [14.9 MB]
Fetched 27.0 MB in 9s (2859 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 50958 files and directories currently installed.)
Preparing to unpack .../00-libdebuginfod-common_0.190-1.1ubuntu0.1_all.deb ...
Unpacking libdebuginfod-common (0.190-1.1ubuntu0.1) ...
Preparing to unpack .../01-libdw1t64_0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libdw1t64:amd64 (0.190-1.1ubuntu0.1) over (0.190-1.1build4) ...

```

```

Preparing to unpack .../00-libdebuginfod-common_0.190-1.1ubuntu0.1_all.deb ...
Unpacking libdebuginfod-common (0.190-1.1ubuntu0.1) ...
Preparing to unpack .../01-libdwt64_0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libdwt64:amd64 (0.190-1.1ubuntu0.1) over (0.190-1.1build4) ...
Preparing to unpack .../02-libelf1t64_0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libelf1t64:amd64 (0.190-1.1ubuntu0.1) over (0.190-1.1build4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../03-libbabeltrace1_1.5.11-3build3_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-3build3) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../04-libdebuginfod1t64_0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.190-1.1ubuntu0.1) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../05-libipt2_2.0.6-1build1_amd64.deb ...
Unpacking libipt2 (2.0.6-1build1) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../06-libsource-highlight-common_3.1.9-4.3build1_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build1) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../07-libsource-highlight4t64_3.1.9-4.3build1_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build1) ...
Selecting previously unselected package gdb.
Preparing to unpack .../08-gdb_15.1-1ubuntu1~24.04.1_amd64.deb ...
Unpacking gdb (15.1-1ubuntu1~24.04.1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../09-libc6-dbg_2.39-0ubuntu8.8_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.39-0ubuntu8.8) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../10-valgrind_1%3a3.22.0-0ubuntu3_amd64.deb ...
Unpacking valgrind (1:3.22.0-0ubuntu3) ...
Setting up libdebuginfod-common (0.190-1.1ubuntu0.1) ...
Setting up libsource-highlight-common (3.1.9-4.3build1) ...
Setting up libelf1t64:amd64 (0.190-1.1ubuntu0.1) ...
Setting up libc6-dbg:amd64 (2.39-0ubuntu8.8) ...
Setting up libdwt64:amd64 (0.190-1.1ubuntu0.1) ...
Setting up libipt2 (2.0.6-1build1) ...
Setting up libbabeltrace1:amd64 (1.5.11-3build3) ...
Setting up valgrind (1:3.22.0-0ubuntu3) ...
Setting up libdebuginfod1t64:amd64 (0.190-1.1ubuntu0.1) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build1) ...
Setting up gdb (15.1-1ubuntu1~24.04.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
vennela_07@VennelaPC:~/week3$ valgrind --leak-check=full ./task5
==2105== Memcheck, a memory error detector
==2105== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==2105== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==2105== Command: ./task5
==2105==

```

```
vennela_07@VennelaPC:~/week3$ valgrind --leak-check=full ./task5
==2105== Memcheck, a memory error detector
==2105== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==2105== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==2105== Command: ./task5
==2105==
Enter command: hello
Stored: hello
Enter command: linux
Stored: linux
Enter command: gcc
Buffer resized to 4.
Stored: gcc
Enter command: exit

Commands stored:
1. hello
2. linux
3. gcc

Memory released successfully.
==2105==
==2105== HEAP SUMMARY:
==2105==    in use at exit: 0 bytes in 0 blocks
==2105==   total heap usage: 13 allocs, 13 frees, 2,176 bytes allocated
==2105==
==2105== All heap blocks were freed -- no leaks are possible
==2105==
==2105== For lists of detected and suppressed errors, rerun with: -s
==2105== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

